

Muonix Detector Prototype Logbook

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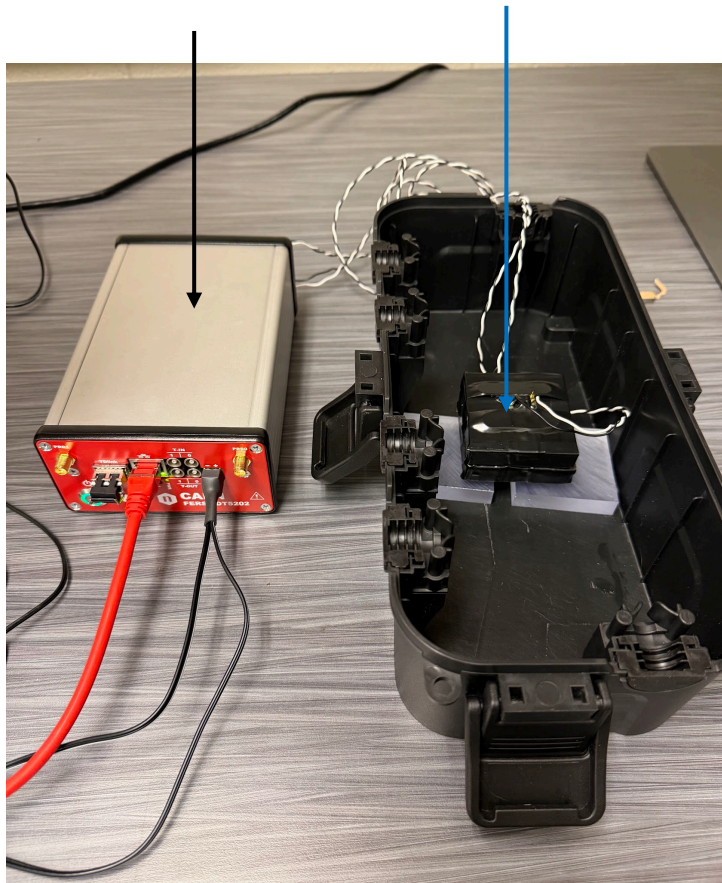
Observing First Muons – Experimental Setup

July 1, 2026

Caen DT5202

64-channel SiPM
readout system

2 × (scintillator + SiPM)



- 2 scintillator+SiPM detectors read out with Caen DT5202
- With 2 detectors stacked on top of each other, we expect to see coincident hits from a muon passing through them
 - **Observe 56 events in 10 minutes → muon rate ~ 0.1 Hz**
- With 2 detectors separated, we expect to see only background
 - Observe 3 events in 10 minutes → ~95% of the events in stacked mode are muons, ~5% are background

Observing First Muons – Results

July 1, 2026

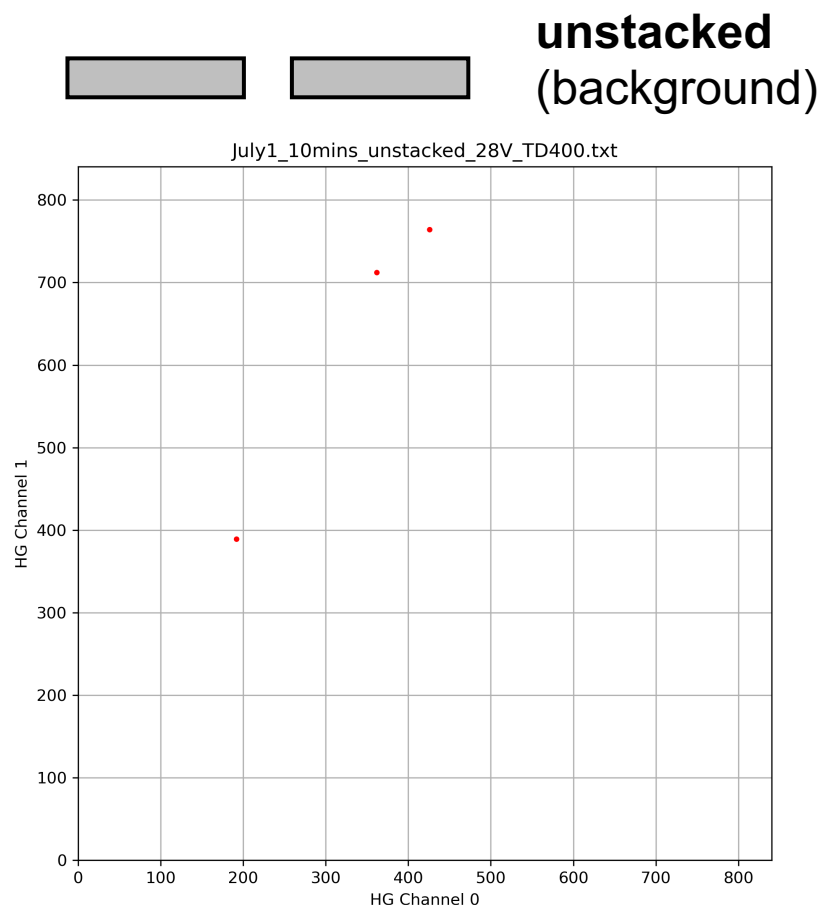
Run for 10 minutes

Plot signals on channel 1 vs. channel

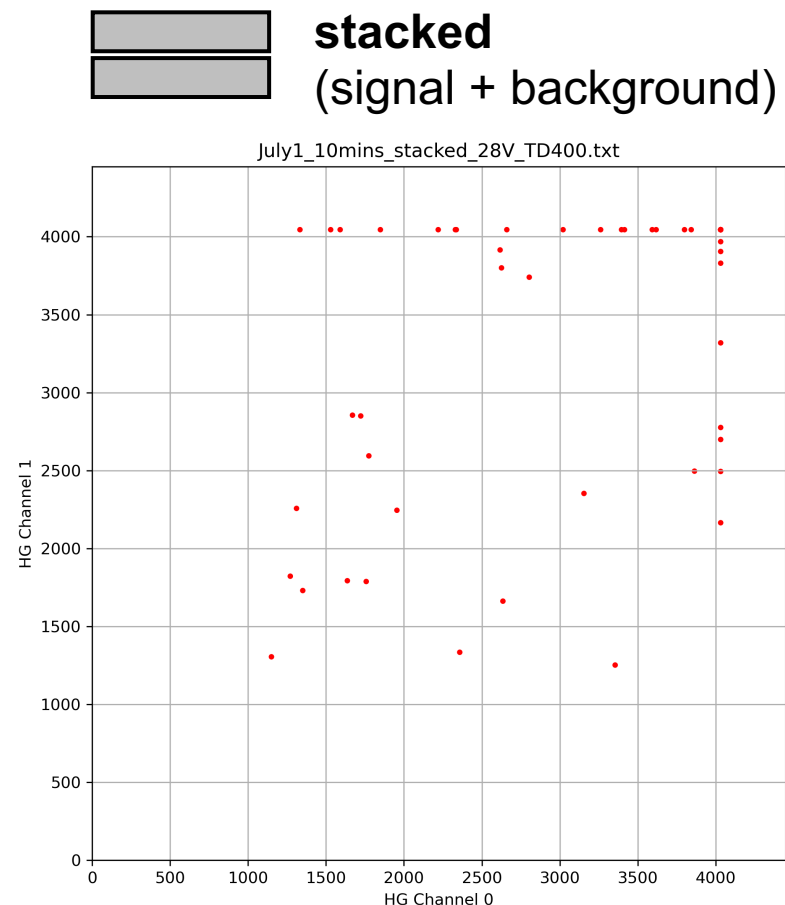
bias voltage 28V, trigger threshold 400

full config at

[July1_10mins_stacked_28V_TD400.conf](#)



3 events



56 events

- **Clear evidence of muons**
- 56 muons / 600 seconds
→ rate ~ 0.1 Hz
- Signal / background ~
53/3 ~ 17:1
- Clear saturation → need
to adjust gain

Observing First Muons – Results

July 1, 2026

Run for 10 minutes

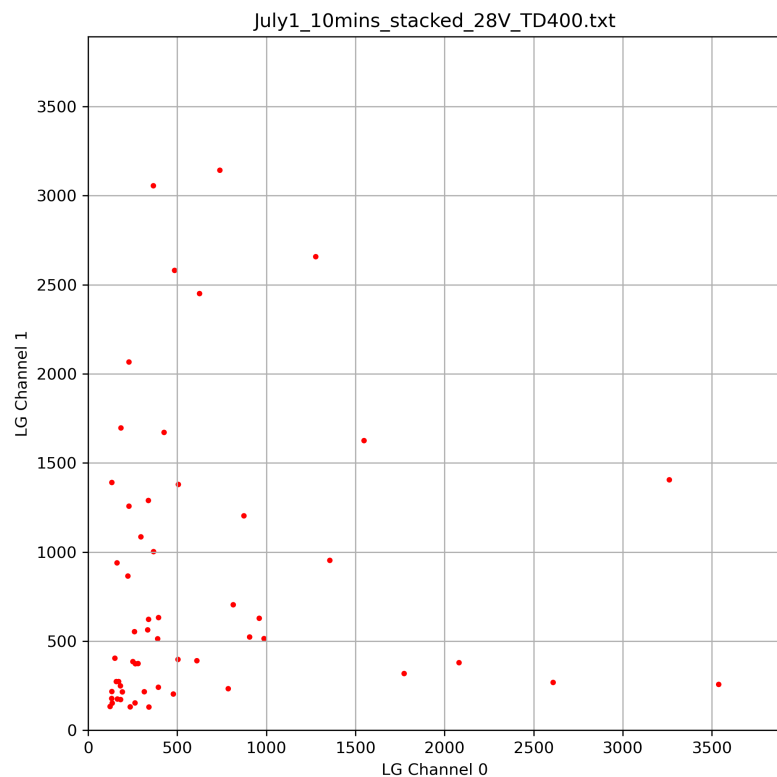
Plot signals on channel 1 vs. channel

bias voltage 28V, trigger threshold 400

full config at

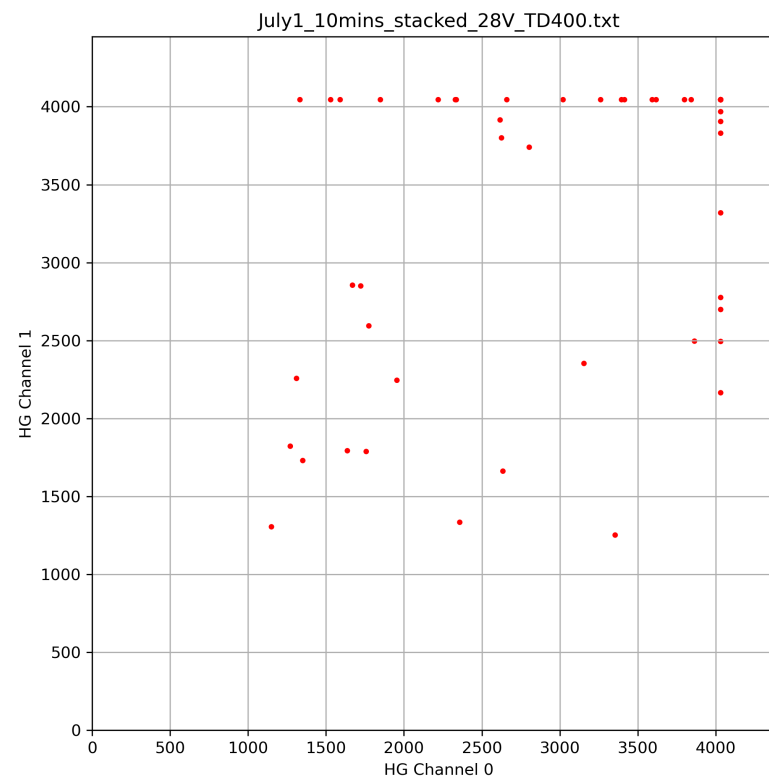
[July1_10mins_stacked_28V_TD400.conf](#)

 **stacked**
(signal + background)



Low gain output

 **stacked**
(signal + background)



High gain output

- Low gain output is not saturating

Observing First Muons – Results

July 1, 2026

Run for 10 minutes

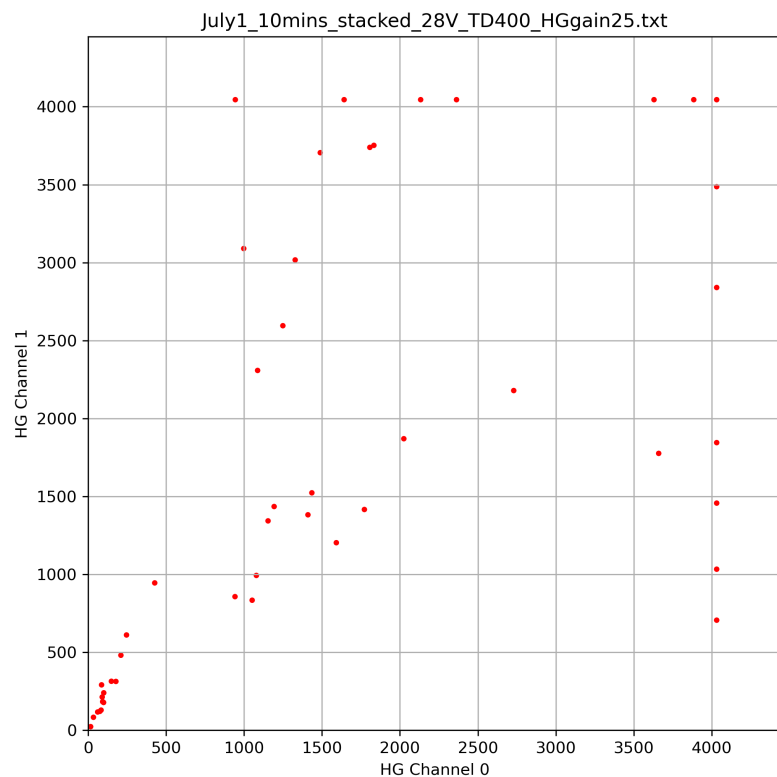
Plot signals on channel 1 vs. channel

bias voltage 28V, trigger threshold 400

full config at

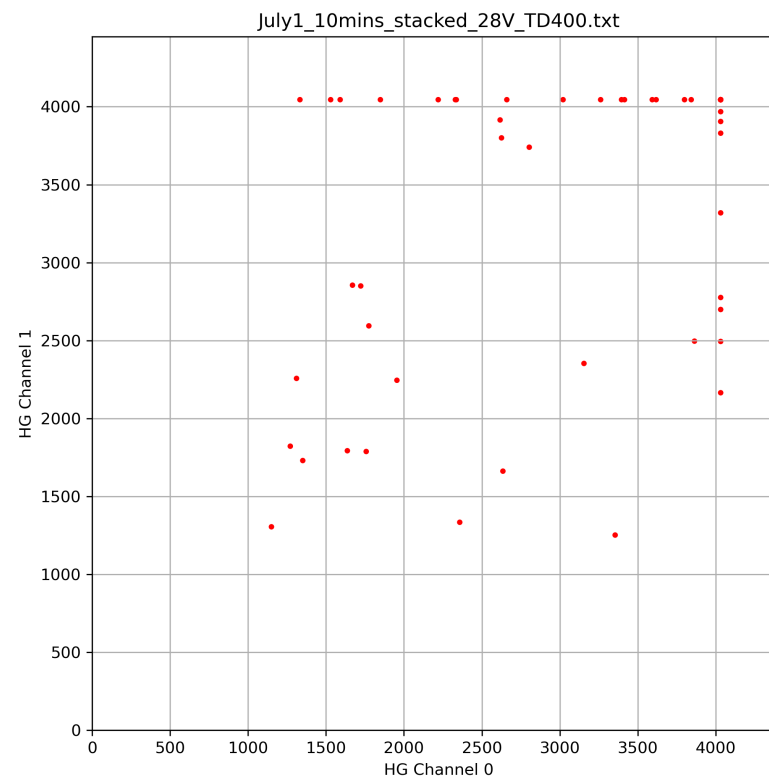
[July1_10mins_stacked_28V_TD400.conf](#)

 **stacked**
(signal + background)



Gain = 25 (47 events)

 **stacked**
(signal + background)



Gain = 50 (56 events)

- Reducing HG gain from 50 to 25 reduces but does not eliminate saturation

Observing First Muons – Results

July 1, 2026

Run for 10 minutes

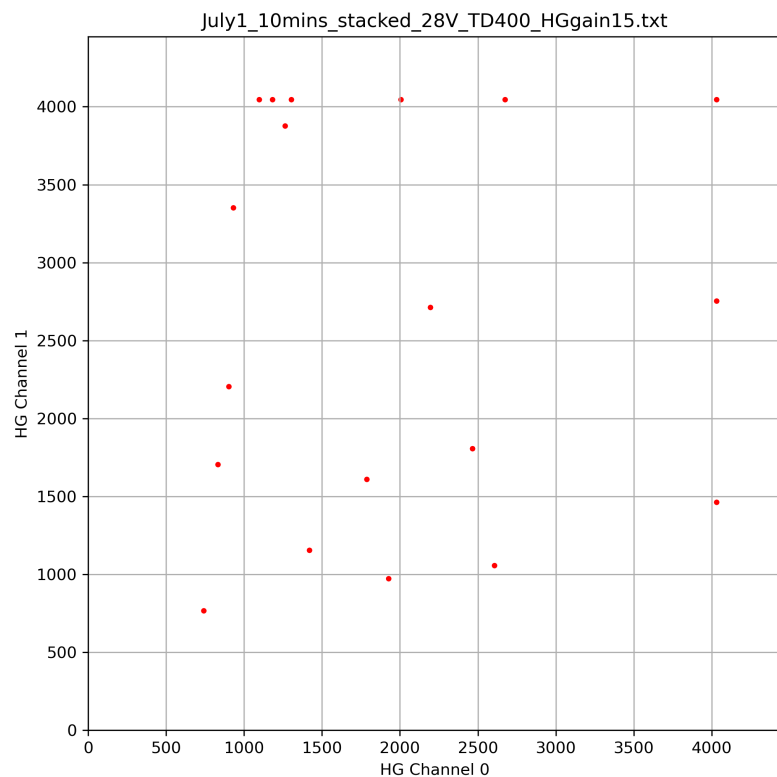
Plot signals on channel 1 vs. channel

bias voltage 28V, trigger threshold 400

full config at

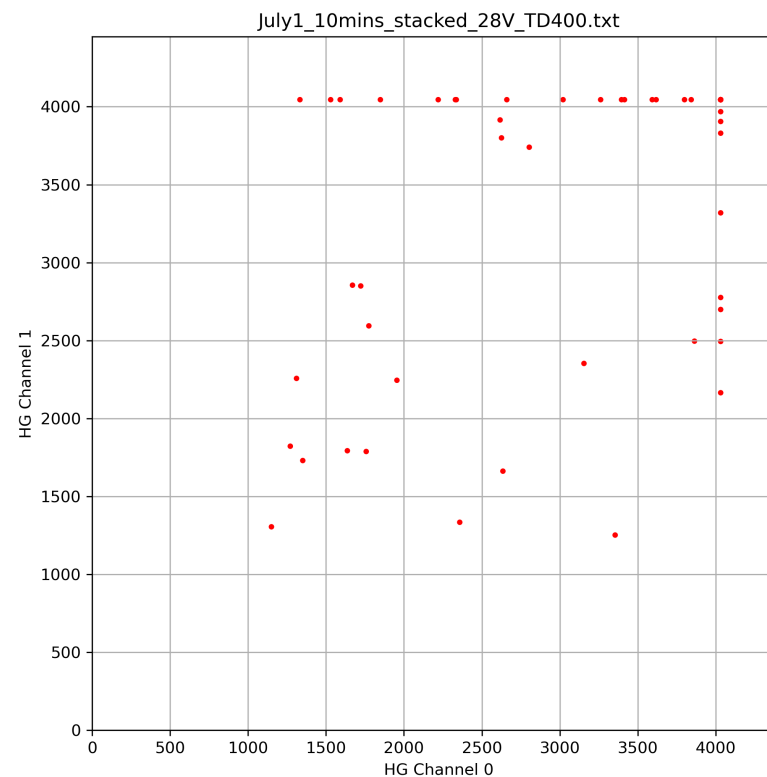
[July1_10mins_stacked_28V_TD400.conf](#)

 **stacked**
(signal + background)



Gain = 15 (19 events)

 **stacked**
(signal + background)



Gain = 50 (56 events)

- Reducing HG gain from 50 to 15 still doesn't eliminate saturation, and the rate has gone down quite a bit